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COMP 4421 (Spring 2026)

Assignment 3

Department of Computer Science and Engineering
The Hong Kong University of Science and Technology

Total = 100 marks

Due: 11:55 pm, May 10, 2026

Assignments must be submitted via CANVAS

Late Policy: Late assignments incur a 10%-score penalty; only one day late is allowed.

Programming Assignment

In this assignment, you should complete the Python functions in the given Jupyter Notebook. Download `programming3.ipynb` and related files from CANVAS.

1. Your submitted Jupyter Notebook must be named `programming3.ipynb`.
2. Runtime requirement: your notebook must finish within 5 minutes.
3. Compress all files into one file named
[your 8-digit student ID]_assign3.rar (or zip)
Example: 09654321_assign3.rar
4. If multiple submissions are made before the deadline, the latest version will be graded.
5. Plagiarism is taken seriously. You may discuss ideas but **must not share code or pseudocode**.
6. Marks will be deducted for requirement violations.

Grading criteria

- Correctness and efficiency of algorithms.
- Adherence to assignment. 20% penalty for any forbidden extra library imports.

Written Assignment

Submit your written assignment as a PDF named

[your 8-digit student ID]_w-assign3.pdf

Example: 09654321_w-assign3.pdf

Include this file in the same compressed submission.

Question W1: Image Transformation

Consider an image I undergoing a transformation to become image I' . The transformation is defined as:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

where (x, y) are coordinates in image I and (x', y') are coordinates in image I' . After the transformation using matrix T , points A, B, C , and D are transformed to points E, F, G , and H respectively.

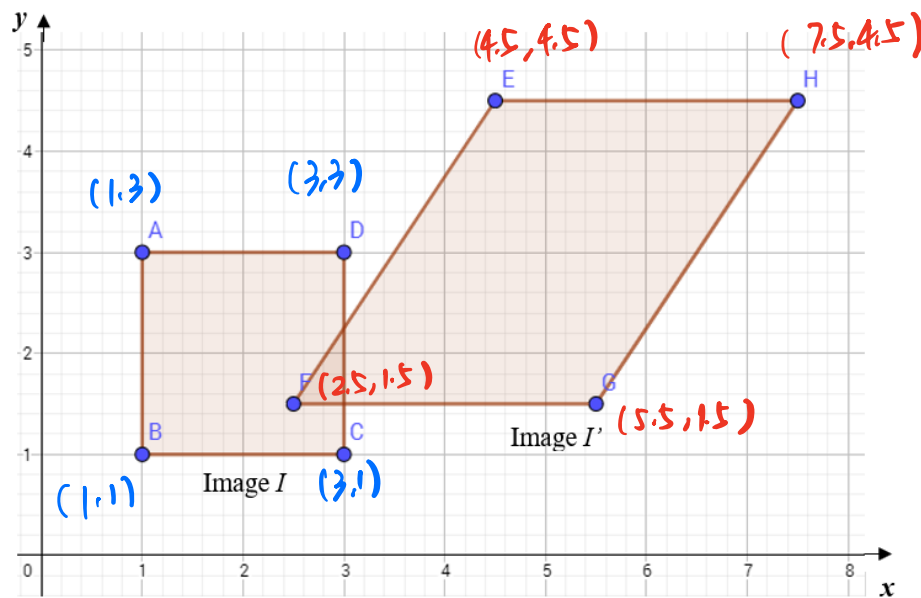


Figure 1: Mapping from Image I to Image I' .

Note: For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Find the transformation matrix $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Please show all your steps clearly.

① $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

② Use $B(1,1) \rightarrow F(2.5,1.5)$ we have:

$$a + b = 2.5$$

$$c + d = 1.5$$

③ Use $A(1,3) \rightarrow E(4.5,4.5)$ we have:

$$a + 3b = 4.5$$

$$c + 3d = 4.5$$

④ Then we get: $b=1$, $a=1.5$, $d=1.5$, $c=0$.

So: $T = \begin{bmatrix} 1.5 & 1 \\ 0 & 1.5 \end{bmatrix}$

Question W2: Quantization

Given a 4×4 gray-level image A as follows:

22	148	31	204
110	180	75	42
63	240	92	15
192	54	10	127

- 2a) Compute the quantization (uniform quantization) of A from 256 levels (8 bits) to 16 levels (4 bits).

$$q = \left\lfloor \frac{x}{16} \right\rfloor$$

$$\begin{bmatrix} 1 & 9 & 1 & 12 \\ 6 & 11 & 4 & 2 \\ 3 & 15 & 5 & 0 \\ 12 & 3 & 0 & 7 \end{bmatrix}$$

- 2b) Compute the column-wise improved gray-scale (IGS) quantization of A from 256 levels (8 bits) to 16 levels (4 bits).

$$s_i = x_i + r_{i-1} \quad q_i = \left\lfloor \frac{s_i}{16} \right\rfloor \quad r_i = s_i \bmod 16$$

$$r_0 = 0$$

$$\begin{bmatrix} 1 & 9 & 1 & 12 \\ 7 & 11 & 5 & 3 \\ 4 & 15 & 6 & 1 \\ 12 & 3 & 1 & 8 \end{bmatrix}$$

Given a 1-D image profile as follows (intensity values range from 0 to 15):

Perform Huffman coding for the image and calculate the compression ratios.

1. Show all your steps clearly.
2. For Huffman coding, show your steps in terms of the tables as shown in the Huffman Encoding part of the lecture notes (Image Compression).
3. You should assign label '1' to the intensity with higher frequency in the Huffman code assignment table.
4. In the calculations of compression ratios, ignore the storage overhead due to the lookup tables, and round your final compression ratio to two decimal places.

Original Source

Symbol	Frequency
2	4
5	3
11	3
3	2
15	2
1	1
10	1

Source Reduction.



Compression Ratios:

Original Size: intensity values range from 0 to 15. $\Rightarrow$ 4 bits each pixel.
Total: $16 \times 4 = 64$ bits.

Compressed Size: $4(2) + 3(2) + 3(3) + 2(3) + 2(3) + 1(4) + 1(4)$
 $= 8 + 6 + 9 + 6 + 6 + 4 + 4$
 $= 43$ bits.

Compression Ratios: $(C_r) = \frac{64}{43} \approx 1.4884$

Round to two decimal places: $C_r = 1.49$